

Method of Zero Voltage Switching (ZVS) and dynamic startup for the Current-fed Push Pull Resonant Inverter

The resonant inverter used in this project relies on high frequency switching to drive an LC tank whose resonant frequency is adjusted using switching capacitors. As tank voltage is sinusoidal and can change phase as capacitance changes, the MOSFETs driving the resonant tank may be forced to turn on with a high drain-source voltage, resulting in significant switching losses and device stress. Zero Voltage Switching (ZVS) is therefore required to ensure each MOSFET switches only at the tank voltage crossing zero voltage, minimising turn-on loss as resonant frequency is adjusted.

Why ZVS is necessary:

1. Reduce switching losses in the inverter:

In a hard switched MOSFET, turn on loss can be approximated by:

$$E_{on} \approx \frac{1}{2} V_{ds} I_d t_{sw} \quad (1)$$

At high operating frequencies, the switching time of the MOSFET becomes small so the losses are dominated by the drain-source voltage and drain current. As the resonant tank generates a large sinusoidal voltage, a MOSFET that turns on at the wrong point in the cycle will lead to increased power dissipation, higher temperature rise and increased EMI (due to fast dv/dt).

ZVS ensures the MOSFET turns on at zero tank voltage and therefore approximately zero drain-source voltage.

2. Impact of frequency control using switched capacitors:

The tanks resonant frequency is:

$$\omega_0 = \frac{1}{\sqrt{L_p C_{eff}}} \quad (2)$$

By switching capacitors in and out C_{eff} varies, this shifts the resonant frequency, the phase of the tank voltage and the timing of the zero-voltage crossings. If they continued switching at fixed frequency or phase after capacitance changes, switching will no longer coincide with tank zero-crossings, causing hard switching, over voltage stress on devices, distorted tank current and failure to re-enter normal resonant operation.

ZVS is essential to re-align switching instants with the tank zero crossings after capacitance change.

As the system continuously adjusts its resonant frequency, fixed timing gate driver signals are not sufficient. Switching must always follow the tank's natural zero crossings to maintain soft switching.

System Overview and Operation

A block diagram showing how the ZVS and startup circuitry can be implemented is shown below:

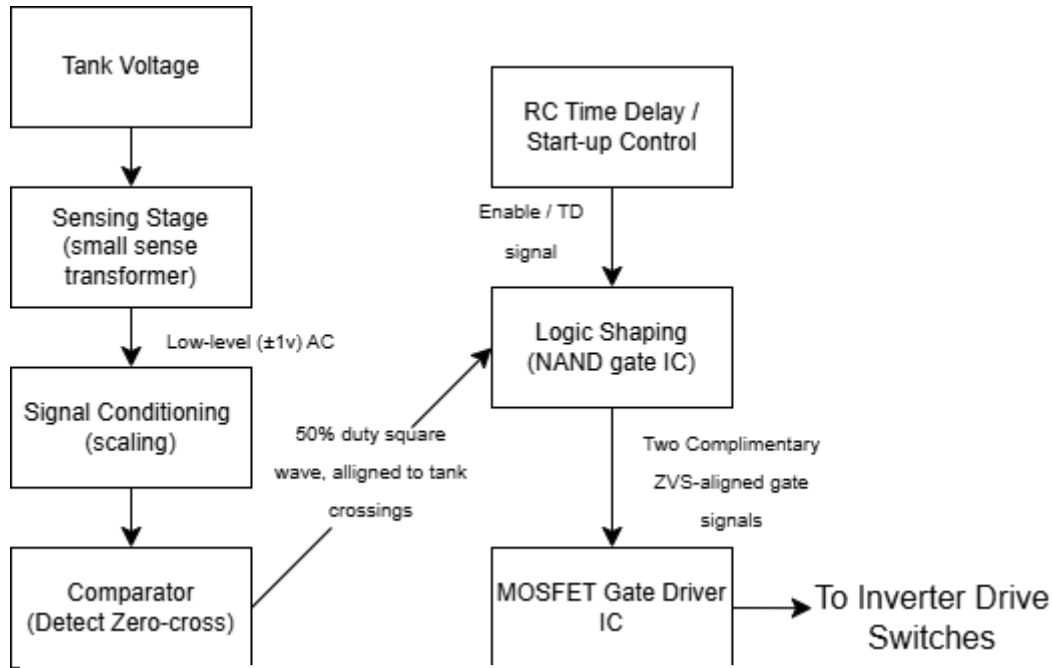


Figure 1: ZVS system block-diagram

The first part of figure 1 involves using a sense (or step-down) transformer across the primary coil. The inductance required for the sense transformer is found from the following equations:

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} \rightarrow N_s = N_p \frac{V_s}{V_p} \quad (3)$$

$$L_s = L_p \left(\frac{N_s}{N_p} \right)^2 \quad (4)$$

By selecting how much voltage should be stepped down, the number of turns of the secondary can be determined to then determine the inductance of the sense transformer coil.

Before the waveform across the sense coil is sampled by the comparator, it is given a DC offset using a simple potential divider network, shown in figure 2 (a). The two resistors, taken of a DC voltage railing, place a bias on the negative terminal of the comparator, hence this node is held at that DC voltage. On the positive terminal the comparator sees the sense coils ac waveform but shifted up by the same bias. Hence the comparator switches polarity as the positive terminal moves around the DC bias, hence the zero-voltage crossing point of the tank waveform, illustrated in figure 2 (b).

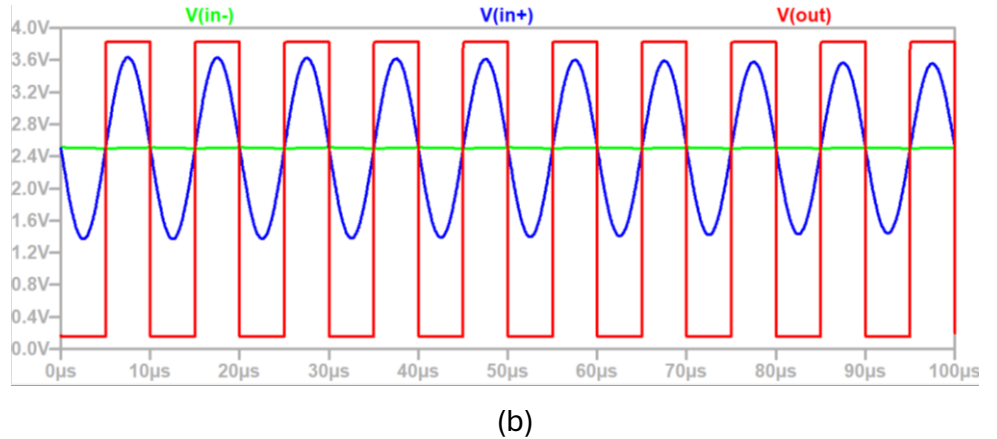
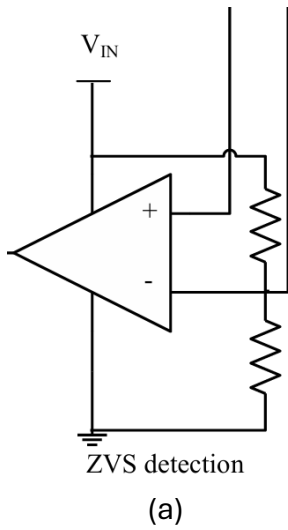


Figure 2: ZVS detection (a) circuit diagram, (b) LTSpice waveforms of comparator negative input (green), positive input (blue) and the output

After the comparator produces the square waveform at the zero crossings a quad NAND gate setup is used to obtain two complimentary square wave signals suitable for driving the push-pull inverter. The basic arrangement is shown in figure 3 (a). One NAND gate is configured as an inverter, producing the logical complement of the comparator output. These two signals – the original and the inverted version – represent the desired conduction intervals for the two MOSFET legs. A second pair of NAND gates is then used to gate these signals with an enable input from the time-delay/start-up circuit, so that switching only occurs when the system has reached a suitable operating condition.

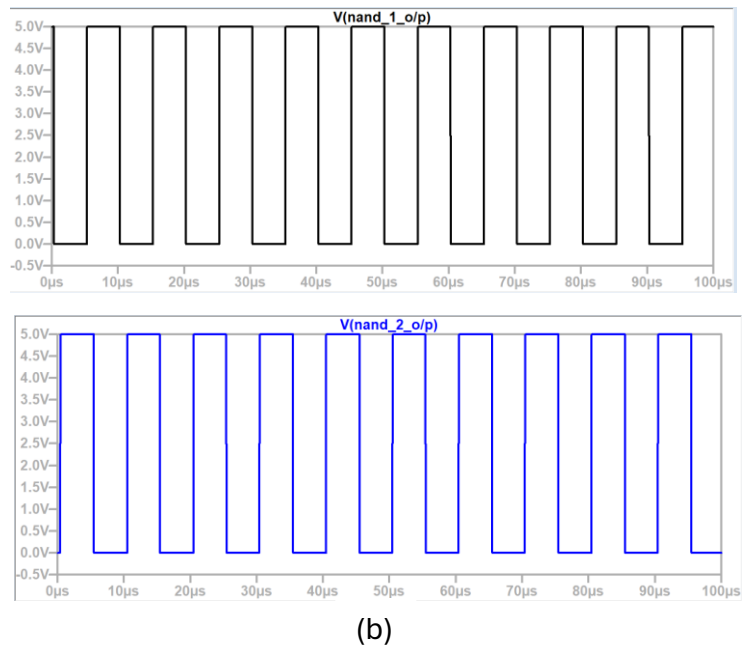
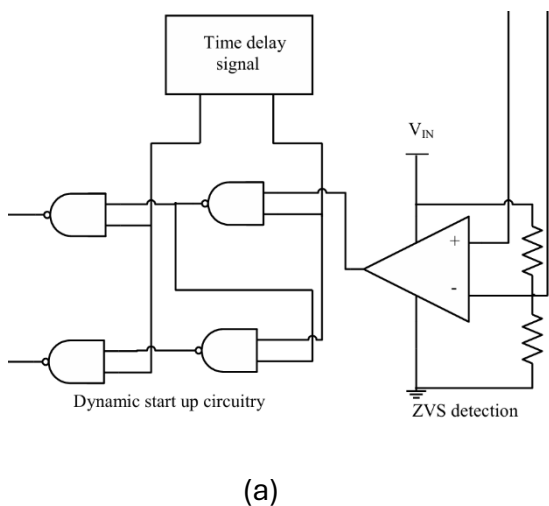


Figure 3: ZVS including NAND gate logic. (a) basic circuit outline, (b) output of top and bottom NAND gates under steady state operation.

To provide the dynamic startup functionality, an RC network is used to generate an enable signal. During startup it is required that both MOSFETs conduct together for a short period of time, this time is given by:

$$t_{short} = I_{int} \frac{L_{dc}}{V_{IN}} \quad (5)$$

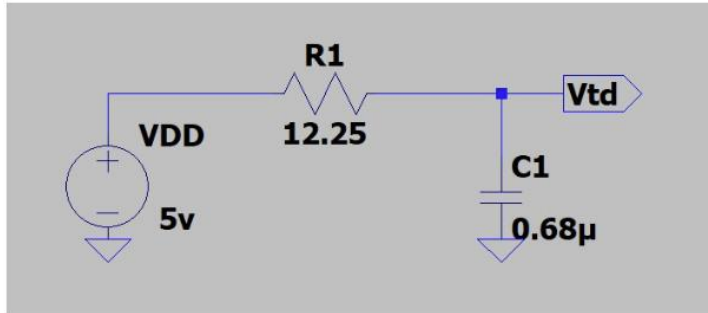
During startup the time delay signal should be low to short both switches, and then high when normal operation begins. To achieve this behaviour a simple RC network, shown in figure 4 (a), can be used. To determine the value of resistance and capacitance required, consider the voltage across the capacitor as it charges:

$$V_C(t) = V_{DD}(1 - e^{-\frac{t}{RC}}) \quad (6)$$

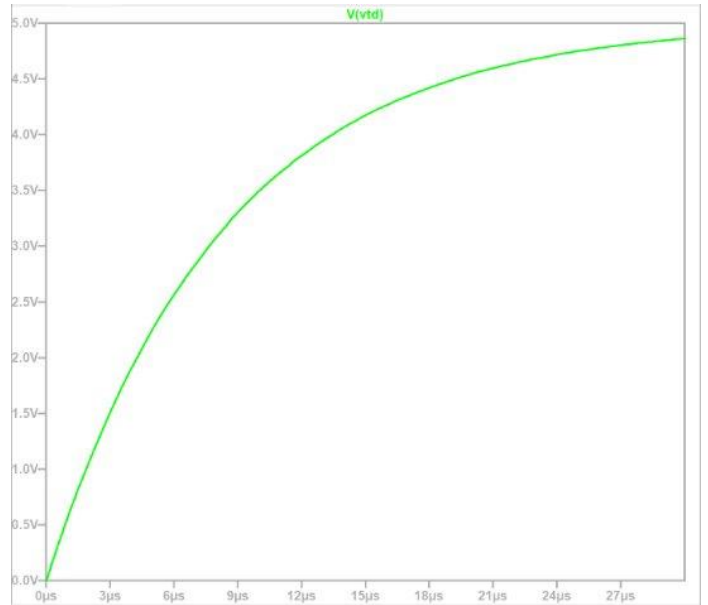
The NAND gates in the chosen IC contain a schmitt trigger input, with hysteresis of $0.7V_{DD}$ or V_{IN} from figure 4 (a). Solving for t in equation 6 where $V_C(t) = 0.7V_{DD}$, gives you the short time of the MOSFETS for the chosen RC:

$$t_{short} \approx -RC \ln(1 - \frac{0.7V_{DD}}{V_{DD}}) \approx 1.2RC \quad (7)$$

Figure 4 (a) outlines an example circuit designed for a short time of $10\mu s$, with figure 4 (b) showing the voltage rise at the time delay voltage node, reaching 0.7 of 5v after approximately $10\mu s$.



(a)



(b)

Figure 4: RC network example to achieve desired time-delay signal. (a) circuit diagram, (b) LTSpice plot of the time-delay signals voltage

The outputs of the NAND logic drive a dedicated MOSFET gate driver IC, which provides the required gate-source voltage swing and peak current. The driver translates the low-level logic signals into fast, rail-to-rail gate pulses with the correct polarity for each

MOSFET. Because these pulses are ultimately derived from the comparator's zero-cross detection, the leading edge of each gate signal is aligned with a zero-voltage crossing of the tank waveform at the corresponding switch node.

Finally, when the effective tank capacitance is changed by switching additional capacitors in or out, the resonant frequency and zero-crossing times shift. However, the sensing transformer and comparator continue to track the new waveform, so the logic automatically realigns the gate signals to the updated zero-voltage instants. In this way the ZVS and start-up circuitry together ensure low-loss, reliable operation of the current-fed push-pull resonant inverter over a range of operating frequencies and tank configurations.